Mzuzu University

Department of Mathematics and Statistics

MATH 1201 Calculus

Test 2

Sunday, March 14, 2021 Time Allowed: 2 hours (13:30-15:30 Hours)

INSTRUCTIONS

- *Check that the paper has **6** questions.*
- *Attempt **all** questions.*
- *Show **all** your working in logical flow and with appropriate notation.*
- *Calculators are not allowed.*

QUESTIONS

1. (a) Differentiate the function $y = \log_5(xe^x)$. [4]

Solution:

$$\begin{aligned} y &= \log_5(x) + \log_5(e^x) \\ \frac{dy}{dx} &= \frac{1}{x\ln 5} + \frac{e^x}{e^x \ln 5} = \frac{1}{x\ln 5} + \frac{1}{\ln 5} \end{aligned}$$

(b) Evaluate the limit $\lim_{x\to 1} \dfrac{x^9 - 1}{x^5 - 1}$. [3]

Solution: $\lim_{x\to 1} \dfrac{x^9 - 1}{x^5 - 1} = \lim_{x\to 1} \dfrac{(x^9-1)'}{(x^5-1)'} = \lim_{x\to 1} \dfrac{9x^8}{5x^4} = \dfrac{9}{5}$

2. Evaluate the limit of the function $\lim_{x\to\infty} \dfrac{e^x}{x^2}$. [5]

Solution: $\lim_{x\to\infty} \dfrac{e^x}{x^2} = \lim_{x\to\infty} \dfrac{(e^x)'}{(x^2)'} = \lim_{x\to\infty} \dfrac{e^x}{2x} = \lim_{x\to\infty} \dfrac{(e^x)'}{(2x)'} = \lim_{x\to\infty} \dfrac{e^x}{2} = \infty.$

3. Find the local maximum and minimum value of the function $f(x) = x^4 - 2x^2 + 3$ [6]

Solution:

$$\begin{aligned} f'(x) &= 4x^3 - 4x = 4x(x^2 - 1) \\ &= 4x(x-1)(x+1) \\ x &= -1, 0, 1 \end{aligned}$$

We can use any of these two methods at this step.

(a) First derivative test.
Hint: We will use the idea that local maximum point is a point where the curve changes from increasing to decreasing and Local minimum is a point where the curve changes from decreasing to increasing. We will make a decision using increasing and decreasing intervals.

Interval	Test Number	$4x$	$x-1$	$x+1$	$f'(x)=4x(x-1)(x+1)$	conclusion
$-\infty,-1$	-2	$-$	$-$	$-$	$-$	Decreasing
$-1,0$	$-\frac{1}{2}$	$-$	$-$	$+$	$+$	Increasing
$0,1$	$\frac{1}{2}$	$+$	$-$	$+$	$-$	Decreasing
$1,\infty$	2	$+$	$+$	$+$	$+$	Increasing

(b) Second derivative test.
$f''(x)=12x^2-4$

Critical Number	$f''(x)=12x^2-4$	Conclusion
-1	$8>0$	Local minimum at -1
0	$-4<0$	Local maximum at 0
1	$8>0$	Local minimum at 1

$f(-1)=2$ local minimum value
$f(0)=3$ local max value
$f(1)=2$ local min value

4. Evaluate the Riemann sum for $f(x) = e^{-x^2}$ for $-2 \leq x \leq 2$ with 4 subintervals using midpoint rule. [8]

Solution: $\Delta x = \dfrac{b-a}{n} = \dfrac{2+2}{4} = 1.$
The values of x are $x_0 = -2,\ x_1 = -1,\ x_2 = 0,\ x_3 = 1,\ x_4 = 2.$
The midpoints are $\bar{x_1} = -\dfrac{3}{2},\ \bar{x_1} = -\dfrac{1}{2},\ \bar{x_1} = \dfrac{1}{2},\ \bar{x_1} = \dfrac{3}{2}.$
The Riemann sum is

$$\begin{aligned} \text{Area}_4 &= \sum_{i=0}^{n-1} f(x)\Delta x = \sum_{i=1}^{4} e^{-x^2} \\ &= e^{-\frac{9}{4}} + e^{-\frac{1}{4}} + e^{-\frac{1}{4}} + e^{-\frac{9}{4}} = 2e^{-\frac{9}{4}} + 2e^{-\frac{1}{4}} \end{aligned}$$

5. Evaluate the derivative of $y = \int_1^{\sqrt{x}} \sin t \, dt$ using Fundamental Theorem of Calculus part 1 (FTC1). [3]

Solution: $= \frac{1}{2\sqrt{x}} \sin \sqrt{x}$

6. Evaluate each of the following integrals.
   (a) $\int (3e^x + 5\cos x - 10\sec^2 x)dx.$ [3]

   Solution: $\int (3e^x + 5\cos x - 10\sec^2 x)dx = 3e^x + 5\sin x - 10\tan x + C$

   (b) $\int_1^e \frac{\ln x}{x} dx$ [5]

   Solution: Let $u = \ln x$, $du = \frac{1}{x}dx$, when $x = 1, u = 0$,$x = e, u = 1$.

$$\int_1^e \frac{\ln x}{x} dx = \int_0^1 u du$$
$$\left[\frac{u^2}{2}\right]_0^1 = \frac{1}{2}.$$

(c) $\int e^{-\theta}\cos 2\theta d\theta$ [7]

Solution: $u = e^{-\theta}$, $du = -e^{-\theta}d\theta$
$dv = \cos 2\theta$, $v = \frac{1}{2}\sin 2\theta$
We have
$\int e^{-\theta}\cos 2\theta d\theta = \frac{1}{2}e^{-\theta}\sin 2\theta + \frac{1}{2}\int e^{-\theta}\sin 2\theta d\theta$

Now we consider $\int e^{-\theta}\sin 2\theta d\theta$.
$u = e^{-\theta}$, $du = -e^{-\theta}d\theta$
$dv = \sin 2\theta$, $v = -\frac{1}{2}\cos 2\theta$
So that we have
$\int e^{-\theta}\sin 2\theta d\theta = -\frac{1}{2}e^{-\theta}\cos 2\theta - \frac{1}{2}\int e^{-\theta}\cos 2\theta d\theta$

Thus,
$\int e^{-\theta}\cos 2\theta d\theta = \frac{1}{2}e^{-\theta}\sin 2\theta + \frac{1}{2}[-\frac{1}{2}e^{-\theta}\cos 2\theta - \frac{1}{2}\int e^{-\theta}\cos 2\theta d\theta]$
$\frac{5}{4}\int e^{-\theta}\cos 2\theta d\theta = \frac{1}{2}e^{-\theta}\sin 2\theta - \frac{1}{4}e^{-\theta}\cos 2\theta + C$
$\int e^{-\theta}\cos 2\theta d\theta = \frac{2}{5}e^{-\theta}\sin 2\theta - \frac{1}{5}e^{-\theta}\cos 2\theta + C_1$ with $C_1 = \frac{4}{5}C$

(d) $\int \sec^5\theta\tan^3\theta\ d\theta$. [6]

Solution:

$$\begin{aligned}\int \sec^5\theta\tan^3\theta\ d\theta &= \int \sec^4\theta\tan^2\theta\sec\theta\tan\theta\ d\theta \\ &= \int \sec^4\theta(\sec^2\theta - 1)\sec\theta\tan\theta\ d\theta\end{aligned}$$

Let $u = \sec\theta$ then $du = \sec\theta\tan\theta$.

$$\begin{aligned}\int \sec^5\theta\tan^3\theta\ d\theta &= \int u^4(u^2-1)\ du = \int (u^6 - u^4)\ du \\ &= \frac{u^7}{7} - \frac{u^5}{5} + C = \frac{\sec^7\theta}{7} - \frac{\sec^5\theta}{5} + C\end{aligned}$$

END OF QUESTIONS